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An enhanced hybrid recommender system using an adaptive optimization approach

Chour Singh Rajpoot[1], Varun Tiwari[1*] and Santosh Kumar Vishwakarma[2]

*Correspondence:
Varun Tiwari
varun.tiwari@jaipur.manipal.edu
[1]School of Computer Science and Engineering, Manipal University Jaipur, Jaipur, Rajasthan, India
[2]Department of Computer Science and Engineering, Gyan Ganga Institute of Technology and Sciences, Jabalpur, India

Abstract
E-commerce platforms generate vast amounts of data, making it challenging for customers to find relevant products efficiently. Recommendation Systems (RS) address this issue by analysing user interactions to suggest personalized items. The two primary RS techniques are Content-Based Filtering (CBF) and Collaborative Filtering (CF). However, CF struggles with tracking changes in user preferences and can perform poorly in sparse datasets. To overcome these limitations, this study proposes a hybrid recommender system that integrates both CBF and CF approaches to enhance recommendation accuracy. The system follows a structured process starting with data collection and preprocessing, which includes data cleaning, discretization, and Principal Component Analysis (PCA) to reduce dimensionality. Feature extraction then captures both content and collaborative features, utilizing methods such as profile construction, content similarity indexing (using Jaccard similarity), neighbor finding (with an improved Manhattan distance refined by Euclidean distance), and item weight generation. For optimizing recommendation ratings, the system employs a novel Adaptive Red Deer Optimization combined with Dandelion Optimizer (ARD-DO). Evaluation metrics such as Accuracy, Precision, Recall, MCC, F-measure, RMSE, MAPE, MSE, and MAE were used to assess performance. Experiments on two datasets show that ARD-DO outperforms other algorithms, achieving minimal MAPE (0.0134 and 0.027186) and RMSE (2.9914 and 2.9225), along with high accuracy (0.9757 and 0.9879). Comparative algorithms like RDO, DLO, BOA, and GFO showed inferior performance, confirming the effectiveness of the proposed hybrid system.

Keywords Recommendation system, Content-based filtering, Collaborative filtering, Red deer optimization, Dandelion optimizer

1 Introduction

Software programs known as recommender systems (RSs) make recommendations for products that a given user is likely to find interesting. When choosing what to purchase, what to put on your playlist, or what to research on the web, these suggestions are used [1]. The design, graphical user interface, and basic recommendation method of the system are tailored to offer practical and efficient recommendations for that particular item. RSs are mostly intended for people who don't have the necessary expertise or experience to assess the deluge of alternatives that are offered on websites [2]. For instance, Amazon.com customizes each customer's experience on its online store using an RS. Customized suggestions are helpful to different people or user populations, although anonymous recommendations are simpler to deliver and usually appear in periodicals or publications [3]. Groups of products deemed to be the finest products or offerings according to the user's interests and constraints are known to be customized recommendations [4]. Usually, when a user does anything like visit a product page, these recommendations are made. Although these suggestions may prove beneficial in some circumstances, the Recommendation Systems (RS) study generally fails to address them. To accomplish this computing work, RSs gather user data, either directly stated as product ratings or deduced from user behavior [5].

The wide range of goods and services that are offered has led to an increase in the popularity of e-commerce websites. However, because there is a lot of data accessible and new services are being added so quickly, customers frequently find it difficult to make the best decisions [6]. The well-being of users has decreased as a result, since having more options can become overwhelming and cause negative emotions. RS has surfaced as a fix for this problem. Employing a range of details and data stored in specialized records about people, available things, and prior interactions, RS generates suggestions for users based on new, unfamiliar items that are relevant to their present work [5, 7]. To facilitate future user-system conversations, users have the option to peruse the recommendations, accept or reject them, and offer input, all of which can be kept in the recommender database. Given the characteristics shared by the examined items, content-based recommendation algorithms are trained to suggest products that are comparable to the user's preferences [8]. These gadgets fit within any of the subsequent groups: demographic, knowledge-based, collaborative filtering, content-based, semantic indexing, and hybrid systems. Content-based methods associate qualities of things with user profiles, while Semantic Indexing uses concepts rather than keywords [9]. While demographic systems promote goods according to user demographics, collaborative filtering makes recommendations based on comparable tastes across users. expertise-based systems, like case-based systems, make recommendations for things based on particular domain expertise [7]. These methods are combined in hybrid systems to overcome drawbacks like unrated items or new-item issues. The benefits of each method can be combined to build an innovative hybrid system by merging these techniques. However, there is a dearth of adequate studies on demographic systems [10].

There are obstacles in RS research when it comes to scenario context, alternative solution comprehension, and user input [11]. Concerns include advantages and disadvantages, impact on user decisions, and determining the worth of recommendations. To create RS systems that work, exploration and exploitation must be balanced [11]. Ongoing research focuses on cross-domain recommender systems, active learning strategies, non-intrusive methods, and merging long- and short-term preferences. Simple goods are usually the focus of RS, while complicated areas call for intricate solutions [12]. Moreover, in ARD-DO, the Red Deer Optimization component performs global population-level exploration, while the Dandelion Optimization is employed on demand for local search depending on recommendation error, rather than using it globally on all candidate solutions uniformly. The adaptive fusion in ARD-DO allows control of hybrid CF-CBF weights and similarity measures to be adaptively controlled with regard to the problem characteristics and computationally efficiently on sparse and evolving data in recommendation tasks. The main contributions of the paper are as follows,

- This paper introduces a hybrid recommender system that combines CBF and CF techniques to enhance user-specific recommendations based on individual preferences and interests.
- The content similarity index utilizes Jaccard similarity to measure the similarity between two sets, effectively handling binary and categorical data. It highlights shared features relative to all in either set, making it suitable for tasks like content similarity and clustering.
- The Neighbour finder uses the Improved Manhattan Distance, a simplified method for high-dimensional space similarity, and the Euclidean distance to capture diagonal relationships and nuanced differences, enhancing accuracy and precision in neighbor-finding tasks.
- The Adaptive Red Deer Optimization with Dandelion Optimizer (ARD-DO) algorithm is used to derive optimal recommendations using extracted features and combines the global search capabilities of Red Deer Optimization with the local refinement strength of the Dandelion Optimizer. This hybrid algorithm enhances rating accuracy by balancing exploration and exploitation in the feature space.

The subsequent sections are arranged as follows: This study's Sect. 2 reviews the literature on approaches used for the RS using Content-Based Knowledge and Collaborating Filtering, and Sect. 3 deliberates the methods for the suggested model. Section 4 presents the paper's outcome and discussion. The research study's conclusion is given in Sect. 5. Table 1 represents the reviews by different authors about the research gaps.

2 Literature review

Keikhosrokiani and Fye [13] introduced a hybrid recommender model for health supplements within an e-commerce context, which leveraged implicit ratings and content similarity to address the cold-start challenge. They extracted features using TF-IDF and matched products with cosine similarity. However, their approach is yet another model that is limited by the static fusion between CBF and CF components. On the other hand, the ARD-DO model enhances the performance and robustness with evolving datasets by using adaptive optimization to dynamically adapt the fusion weights between CB and CF methods.

Parthasarathy and Sathiya Devi [14] presented the development of a new state-of-the-art hybrid RS that will offer its users, through preference, recommendations. It pointed out the failures of the various RS in existence and presented another approach that integrated CBF and CF techniques. To deliver the greatest potential recommendation ratings, the FireFly algorithm with Weighted Crow Search Algorithms was used to collect data from user profiles and items.

Jomsri et al. [15] developed a hybrid model for digital libraries, combining CF and CBF to improve recommendation relevance. The hybrid model, developed by Jomsri et al., attempted to improve the relevancy of the recommendations; though it is limited by data sparsity and lacks a generalizable, scalable solution for very large datasets. The ARD-DO method optimizes the hybrid fusion by avoiding sparsity in the traditional recommendation approach by using adaptive optimization methods, allowing for efficient adaptive

Table 1 Comparative summary of various RS

Ref no	Year	Method used	Strengths	Limitations
[16]	2024	Hybrid model (CF + CBF) for health supplements recommendation	Combines implicit ratings and content similarity Addresses cold-start issues	Static fusion between the CBF and CF components Limited scalability and adaptability in dynamic datasets
[17]	2023	Hybrid model (CF + CBF) with FireFly algorithm and Weighted Crow Search Algorithm	Integrates advanced optimization algorithms Improves recommendation ratings accuracy	Complexity of algorithm integration- Does not address real Time adaptation to changing user preferences
[18]	2024	Hybrid model (CF + CBF) for digital libraries	Combines CF and CBF to improve recommendation relevance Applicable to large datasets	Struggles with data sparsity Lacks a scalable solution for very large datasets
[19]	2024	Hybrid model (NCF + RNN + CF)	Leverages both latent user-item interactions and sequential patterns- Deep learning The base model provides improved accuracy	High computational cost due to deep learning models Struggles with data sparsity
[20]	2025	Neural Collaborative Filtering (NCF)	Captures non-linear relationships between users and items Outperforms traditional models like MF and CBF in accuracy	Requires large datasets- Computational complexity Cold-start problem persists
[21]	2024	Fusion recommendation using frequent itemset mining	Improves accuracy and handles sparse data	-Limited in capturing deep user–product relationships
[22]	2023	Hybrid model (CF + CBF)	Improves diversification and accuracy of recommendations Addresses cold-start and diversity issues in recommendations	Lacks sufficient real-time adaptability Scalability could be limited in massive datasets
[23]	2024	Hybrid model (CF + CBF) with dynamic fusion based on similarity measure, metadata analysis, and hybrid scoring functions	Dynamic fusion of CF and CBF allows for adaptive recommendations- Performs well in new user and new item scenarios Enhances personalization	Static hybrid fusion methods can be limited for highly dynamic datasets Potential challenges in fine-tuning hybrid fusion for specific cases
[24]	2024	Hybrid model (CF + CBF) for personalized product recommendations	Solves sparsity and adapts to user preference dynamics Can recommend for both new and returning users	Static fusion between CF and CBF could limit dynamic adaptability Relies heavily on user profiles
[25]	2024	Hybrid model (CF + FSA) for e-learning recommendations (CFISAR)	Uses fine-grained sentiment analysis (FSA) to improve recommendations- Dynamic updates to learner profiles Personalized and evolving content	Could struggle with large-scale real-time updates Computational demand of continuously integrating FSA for dynamic content
[26]	2024	univariate filters (Relieff, Pearson Correlation, Mutual Information, ANOVA, Chi2) and multivariate filters (CFS, DISR, CON) on classifiers (MLP, SVM, XGBoost, Random Forest)	Efficient feature reduction Considers feature dependencies To improve classifier performance	Univariate filters ignore feature interactions Multivariate filters are more computationally complex Classifier performance depends on feature quality and parameter tuning
[27]	2024	Filter-based FS + Interpretability (Permutation Feature Importance, PDP, Global Surrogate) on CICIDS2017, CICIDS2018, CIC-ToN-IoT	Enhanced interpretability of black-box IDS models Global Surrogate balances accuracy–interpretability trade-off Consistency across multiple interpretability techniques	Added an interpretability layer increases computational overhead Limited scalability across very large datasets

Table 1 (continued)

Ref no	Year	Method used	Strengths	Limitations
[28]	2025	CTGAN-ENN + FS (filters + wrappers) + Interpretability (SHAP, LIME, GS)	Superior performance compared to SMOTE, ADASYN, WGAN Accuracy > 99% across multiple datasets Provides both local and global interpretability	GAN-based models require careful tuning Synthetic data generation may introduce bias if not balanced properly
[29]	2025	Ensemble FS (intersection, multi-intersection, fusion) with RF, XGB, MLP, SVM	Fusion and multi-intersection filters outperform singles RFE and Boruta are effective with ensemble classifiers GS interpretability resolves the accuracy–interpretability trade-off	Wrapper ensembles are less effective than single wrappers Increased complexity in ensemble FS strategies
[30]	2025	FS (Consistency, Pearson, DISR, Chi2 + Wrappers: Boruta, BorutaShap, RFE, GA) with RF, XGB, MLP, SVM on CICIDS2018	RFE + CON with ensemble models reduces attributes without affecting classification Improved interpretability with FS	Wrapper methods computationally expensive Limited generalization beyond the CICIDS2018 dataset
[31]	2024	CTGAN + CON FS with RF, XGB, LGBM, DNN on CICIDS2018	XGB with CTGAN achieved 99.99% accuracy using only 13 features Outperformed SMOTE and ADASYN Demonstrated efficiency of synthetic data + FS	Requires ≥ 70% synthetic data for optimal performance GAN training complexity and stability issues

scalable recommendations, while also improving relevancy in recommendations by dynamically adjusting weights.

Sami et al. [16] developed a hybrid recommender system that integrates Neural Collaborative Filtering (NCF), Recurrent Neural Networks (RNNs), and Collaborative Filtering (CF) to leverage both latent user-item interactions and sequential patterns of user behavior. Although the approach yielded promising results, the performance is limited by data sparsity and would require significant computational resources for training. In contrast, Aids to Decision-Making - a Decision Optimization (ARD-DO) is a more computationally efficient alternative approach to hybrid recommender systems and does not rely on deep learning; rather, it relies on adaptive optimization to improve hybrid recommendations by reducing computational load while maintaining prediction accuracy.

Efrizoni et al. [17] proposed a Neural Collaborative Filtering (NCF) approach to improve recommendation performance by capturing the non-linear relationships between users and items. Their method outperformed traditional models such as Matrix Factorization (MF) or Content-Based Filtering (CBF) by capturing nonlinear relationships between users and items. Nonetheless, NCF models have issues relating to computational complexity and cold-start difficulty, including the requirement to be trained with large data sets.

Kang and Wang [18] proposed a novel hybrid recommendation algorithm that utilizes frequent itemset mining to improve personalized product recommendations on e-commerce markets. Their algorithm addresses data sparsity, cold start, and redundancy problems by compressing the commodity dataset and discovering frequent itemsets, while using the ranked user–commodity interests to establish similarity. This allows the algorithm to be dynamic in terms of user preference and, therefore, increases its accuracy and efficiency in the recommendation process. Their competitive evaluations showed improvements over standard data mining algorithms in terms of reduced computation time and increased accuracy, leading to more accurate and responsive e-commerce recommender systems.

Widayanti et al. [19]. proposed an RS hybrid technique, which incorporated CBF and CF. This technique improved the diversification and accuracy of the set of recommendations by designating the inadequacies of different approaches. Their study focused on data collection, storage management, and the CF and CBF techniques. This hybrid model upgraded the system's understanding of the user's preferences by the system and provided more intelligent and personalized recommendations. This innovation in the domain of RS has ramifications for more efficient algorithms.

Sarkar et al. [20] developed a hybrid RS that combines Collaborative Filtering (CF) and Content-Based Filtering (CBF) to address common challenges in RS, such as cold-start and data sparsity. They dynamically reconcile CF and CBF contributions using a similarity measure, metadata analysis, and hybrid scoring functions, resulting in more personalized and diverse recommendations. The hybrid model performed exceptionally well in new user and new item scenarios and mitigated the potential complication of changes in user behavior, which resulted in a more adaptive, dynamic, and richer user experience in the advanced procesA more clarified methodology for data-driven personalized services contributed to modern digital platforms.

Patil et al. [21] suggested a hybrid RS that merges Collaborative Filtering (CF) and Content-Based Filtering (CBF) methodologies to respond to problems of sparsity and dynamics of user preferences. The hybrid algorithm will provide personalized product recommendations to both new and returning users by utilizing user profiles, which are based on the observation of prior purchases and feedback responses.

Alatrash et al. [22] proposed a new e-learning RS that combines Collaborative Filtering Models (CFMs) with fine-grained sentiment analysis (FSA) to enhance recommendations based on learner profiles. The proposed framework, called CFISAR, is designed to provide dynamic, personalized recommendations by continuously refining the learner's profile and delivering content tailored to their evolving interests. The CF component builds profiles of the learners based on preferences, while the FSA improves the profile by looking at reviews and ratings given by users. Additionally, the system is dynamic, regularly adding new content to the system and predicting ratings for new items. This means the recommendations will remain useful and fresh. This study of recommendations has improved prediction accuracy and user personalization for e-learning sites, while also adapting to user needs over time.

Zouhri et al. [23] analyzed the effect of five univariate filters (Relieff, Pearson Correlation, Mutual Information, ANOVA, Chi2) along with three multivariate filters (CFS, DISR, CON) on four classifiers (MLP, SVM, XGBoost, Random Forest) for the classifiers on the CIC-IDS2017, CSE-CIC-IDS2018, and CIC-ToN-IoT datasets. High-dimensional datasets can lead to overfitting and reduce classifier performance. The outcomes convey that the use of CON and DISR, among the multivariate filters, helps to effectively diminish the features along with the performance on the detection task.

Zouhri et al. [24] designed an interpretable ML-based IDS that utilizes filter-based FS techniques on three datasets; they built 18 different versions of the black-box model and applied interpretability techniques, including Permutation Feature Importance, Global Surrogate, and Partial Dependence Plot for interpretation of their model. They report that the use of Global Surrogate as an interpretation method provided comparable levels of accuracy and interpretability among all opaque models analysed.

Zouhri and Idri [25] presented CTGAN-ENN, a novel hybrid framework that integrates Conditional Tabular GANs with Edited Nearest Neighbor (ENN) for undersampling in addition to enhancing both minority class representation and interpretability utilizing SHAP, LIME, and Global Surrogate techniques. CTGAN-ENN consistently outperformed traditional oversampling techniques (SMOTE and ADASYN) by achieving accuracy rates on the CICIDS2018, CIC-ToN-IoT, and NF-UNSW-NB15-v2 datasets.

Zouhri and Idri [26] evaluated eight filter-based and four wrapper-based FS techniques across four classifiers and four intrusion datasets. They introduced ensemble FS strategies (intersection-based, multi-intersection, fusion-based) and found that fusion and multi-intersection filters yielded superior performance. Recursive Feature Elimination and Boruta were particularly effective when paired with ensemble models like RF and XGB. The Global Surrogate method again proved valuable in resolving the accuracy–interpretability trade-off.

Zouhri and Idri [27] assessed FS techniques on the CICIDS2018 dataset, finding that combining RFE and CON-based FS with the use of ensemble classifiers identified a reduced number of features while still achieving comparable classification performance, indicating that FS is critical to the simplification of IDS models.

Zouhri and Idri [28] introduced the CTGAN with FS based on CON on CICIDS2018 provided better performance than using either SMOTE or ADASYN in generating balanced training data. The XGB classifier produced an accuracy of 99.99% using only 13 features from a total of 81 features, which demonstrates that the generation of synthetic data, when combined with efficient FS techniques, can yield high-performing and readily interpretable IDS. Table 1 shows the comparative analysis of existing RS.

2.1 Research gap

While hybrid recommender systems that combine Collaborative Filtering (CF) methods and Content-Based Filtering (CBF) methods have shown great success in recommending accuracy, they face many challenges in the face of the subsequent cold-start problem as well as data sparsity, even if the recommendations can be personalized. This portion also suffers from static fusion within the hybrid CF-CBF model. Hybrid recommender systems incorporating static fusion weights between CF and CBF components, incorporating the approaches shown in the proposed models of Keikhosrokiani and Fye (2024) and Sarkar (2025), leave the model less applicable to dynamic user preferences and the development of changing datasets. There has been progress with both deep NLP-based approaches, which notoriously yield inaccurate reusable datasets due to poor/no scalability (Neural Collaborative Filtering (NCF) based approaches) or additional externally-applied heuristics in a neural-based architecture. Adaptive agents, as seen in Pham et al. (2023) and Alatrash et al. (2024), have made progress in the literature within adaptive optimization to solve uncertainty in personalization. And while both approaches can increase an adaptive measure, both claim challenges regarding real-time adaptation as well as preserving computational costs. The ARD-DO (Adaptive Red Deer-Dandelion Optimization) model aims to fill the desired gaps above by proposing dynamic weights adjusting the CF and CBF components in CF-CBF recommender systems in real-time. When user feedback comes into play, this affords its place as either more scalable, safe, and computationally cheap or, equally beneficial by yielding improved user satisfaction and relevant recommendations, all without needing an extensive dataset or an expensive neural deep-learning model.

3 Proposed methodology

The study employs multiple techniques for data collection, preprocessing, feature extraction, and recommendations. In the preprocessing step, techniques such as Principal Component Analysis (PCA), data discretization, and data cleaning are incorporated. Once principal component analysis is initiated, the system can process collected data and utilize Neighbor Finder by combining the Manhattan Distance with the Euclidean Distance to optimize and join user dimensionality together to accurately measure the similarity of users in dimension and generate users' recommendations in a user's closest to themselves. Neighbors Finder uses the Jaccard Similarity Index for content similarity and the Improved Manhattan Distance and Euclidean Distance to improve the recommendation neighbor finder approach. Finally, the optimized recommendation is generated using ARD-DO (Adaptive Red Deer-Dandelion Optimization). The overall architecture of the system is depicted in Fig. 1.

3.1 Dataset description

This research utilizes two main types of data: medical recommendations and product recommendations, to explore the efficiency and generalizability of the recommended hybrid recommender system (CBF + CF + ARD-DO). Each type combines two publicly available benchmark datasets to ensure diversity with regard to the domain and sparsity.

- **Medical Recommendations**.

Hospital rating dataset [29]: This dataset was sourced from the Centers for Medicare & Medicaid Services (CMS) and comprises data from more than 4,000 Medicare-certified

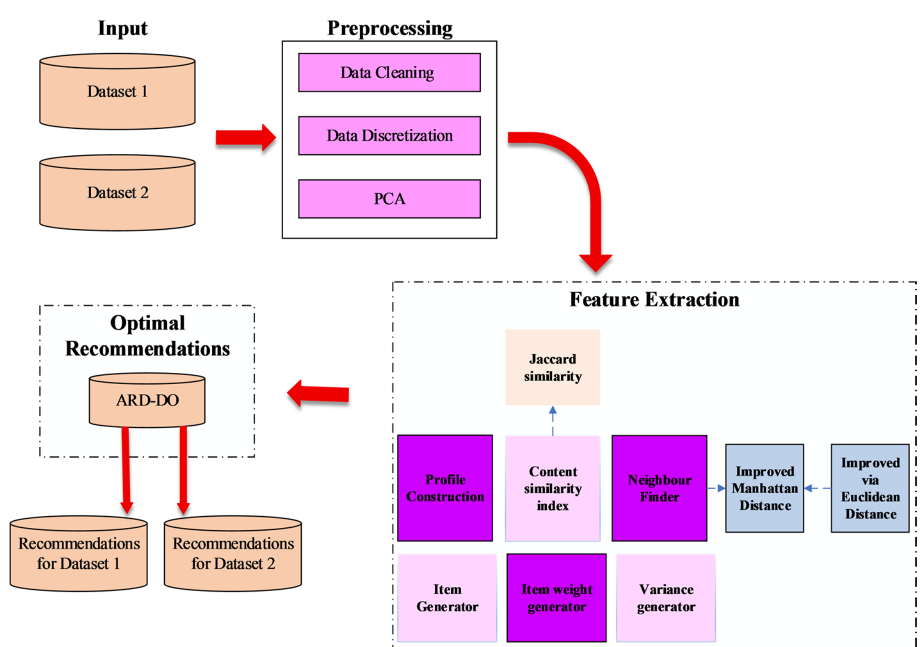

Fig. 1 Overall Architecture of the proposed model

hospitals in the United States. It contains roughly 50,000 total records that include the hospital name, hospital rating, hospital location, and various measures of service quality. The data is structured with a combination of numerical and categorical data types, which makes it suitable for providing healthcare recommendations.

Medical recommendations dataset [30]: A custom dataset designed for medicine recommendations based on patient symptoms and diagnosis patterns. It contains approximately 2,500 patient profiles, where each profile contains features such as symptoms, illnesses that appear, and recommended treatments. The dataset has moderate sparsity (≈ 0.28 density) and can be used to test the recommendations for health-based personalized recommendations.

- **Product Recommendations**.

Amazon product analysis & recommendation system [31]: This data set contains around 1,000 unique products with over 10,000 user-generated reviews that have attributes such as product title, product category, price, and rating. It is a dense consumer dataset, which presents an ideal testing case for evaluating recommendation quality against e-commerce products.

Recommendation system for Amazon products [32]: A large-scale dataset consisting of over 100,000 transactions/user-item interactions that was primarily used by Amazon for item-to-item collaborative filtering. By design, the dataset is sparse (≈ 0.12 density), providing an ideal testing case for cold-start and scalability analysis.

3.2 Pre-Processing

The datasets collected will go through a data Pre-Processing phase comprising a series of processes: data cleaning, data binning, discretization, and Principal Component Analysis (PCA), etc. This is done to improve the quality of the data and to prepare it for proper recommendation generation. Data pre-processing ensures that the data is as consistent and noise-free as possible in order to prepare the data for feature extraction and optimization.

3.2.1 Data cleaning

The purpose of data cleaning is to remove missing values, duplicates, and inconsistencies to ensure that the data is reliable and consistent for the merged datasets. Data cleaning will involve conducting imputation for missing values, removing outliers, and transforming categorical data into a consistent format. The purpose of data cleaning is to ensure better data quality and integrity in the data, so that the data can be used for further analysis. Furthermore, missing values were handled, and categorical variables were encoded.

3.2.2 Data discretization

Data discretization is a process of transforming continuous attributes into categorical attributes of selected intervals or specific attributes. Data discretization approaches, like equal-width binning or equal frequency binning, are used to consolidate values into representative intervals or categories. In general, data discretization helps reduce data complexity, facilitates more interpretable models, and results in little information loss. Continuous features were transformed into discrete intervals for better interpretability.

3.2.3 Principal component analysis (PCA)

PCA is applied for dimensionality reduction to delete redundancy while producing features that are the most informative. PCA restructures the structure of correlated attributes into a smaller set of uncorrelated principal components, while maintaining the variance in the data. PCA is documented as producing augmented analysis; therefore, it is carried out only after presenting all data, and is carried out on standardized data, which normalizes the scales of attributes. The reduced features collected from PCA will reduce when used in computation time, while also providing greater precision in our hybrid recommender system.

3.3 Feature extraction

After preprocessing, the system moves to feature extraction to produce a meaningful representation of users and items for hybrid recommendation. This step extracts content-based (CB) and collaborative filtering (CF) features simultaneously to improve personalization, similarity estimation, and prediction accuracy. Feature extraction includes building user profiles and item profiles, measuring similarities, finding neighbours, generating candidate items, assigning item weights, and analysing variance. Figure 2 shows the flowchart of the feature extraction process in the proposed RS.

3.3.1 Profile construction

User Profiles and Item Profiles are developed to characterize critical features of items and user preferences, respectively. An excerpt of the profile of the individual item is created using keywords and attributes provided by the seller, while a user profile is formed based on the aggregation of item features from the items a user has interacted with or rated. The profiles are used as a basis for computing similarity measures in subsequent stages.

3.3.2 Content similarity index

The content similarity index determines users or items that have similar characteristics based on their profile vectors. In the proposed system, both cosine similarity and Jaccard similarity are employed, as they capture complementary aspects of similarity. Cosine similarity is used to measure the similarity between weighted profile vectors, where feature importance and magnitude are relevant. It is defined as Eq. 1.

$$UserCS\left(User_1, User_2\right) = Cosine\left(\overline{User_1}, \overline{User_2}\right) = \frac{\overline{User_1}.\overline{User_2}}{\left|\overline{User_1}\right| \times \left|\overline{User_2}\right|} \tag{1}$$

Here, $User_1$and $User_2$ represent user rating vectors, "·" denotes the dot product, and $\mid \cdot \mid$ indicates vector magnitude. To evaluate item-level overlap, the Jaccard similarity is indicated in Eq. 2.

$$Sim(w,v)^{jacc} = \frac{\mid i_w \cap i_v \mid}{\mid i_w \cup i_v \mid} \tag{2}$$

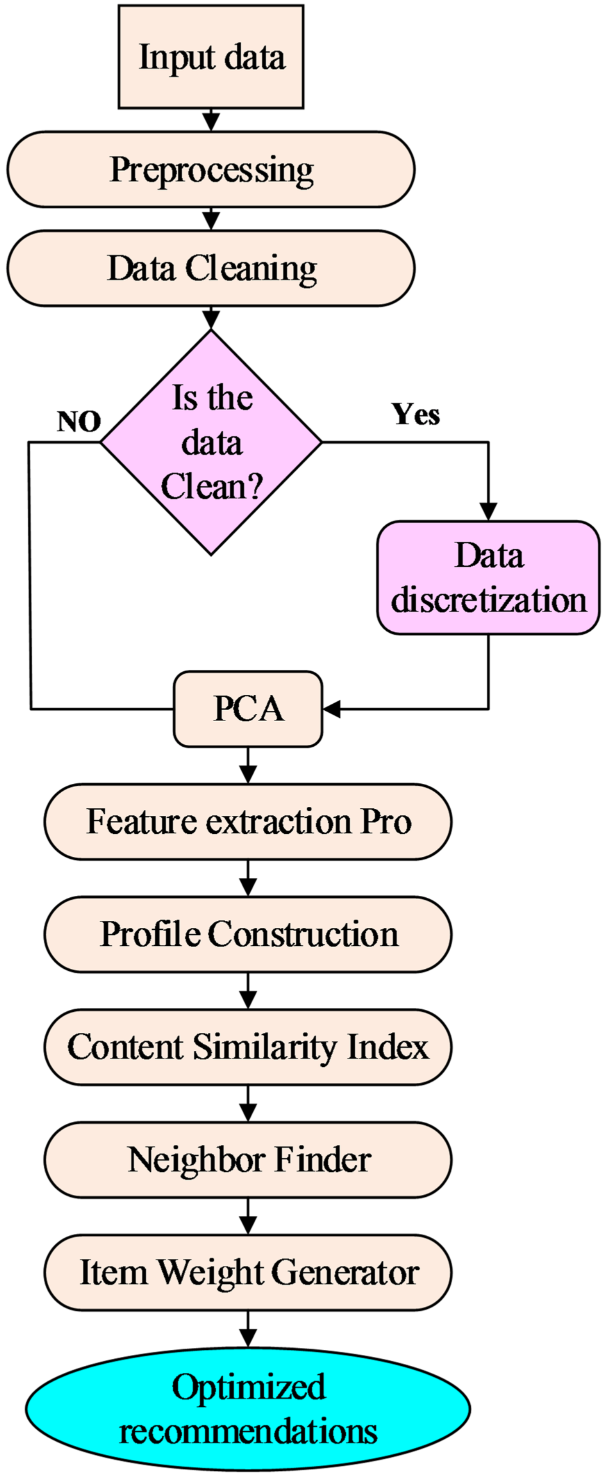

Fig. 2 Flowchart of the feature extraction in the proposed RS

where i_w and i_v denotes the sets of items rated by users $w\ and\ v$, respectively; $\cap$ represents the intersection, $\cup$ represents the union. In contrast, Jaccard similarity is utilized to measure the commonality between binary or categorical features, for example, common objects or attributes, and is more effective on sparse data. With the combination of both cosine similarity measures on continuous features and Jaccard similarity on common categories, the proposed system enhances the accuracy level in measuring content similarity.

3.3.3 Neighbour finder (NF)

The NF determines the closest users to the target user based on a combination of rating behaviors and content similarity. NF begins by using the Pearson correlation coefficient to evaluate linear relationships across users' rating behavior in Eq. 3,

$$UserRS\left(User_1, User_2\right) = \frac{\sum_{a\in i_{1,2}}(r_{a,user_1} - ar_{user1})(r_{a,user_2} - ar_{user2})}{\sqrt{\sum_{a\in i_{1,2}}\left(r_{a,user_1} - ar_{user1}\right)^2 \left(r_{a,user_2} - ar_{user2}\right)^2}} \quad (3)$$

Where, $i_{1,2}$ is the set of items rated by both users, $r_{a,user_i}$ is the rating of the item a by user i, and r_{user1} is the average rating by the user i. To refine similarity estimation, the system applies an Improved Manhattan Distance combined with the Euclidean Distance, which enhances the geometric accuracy of user proximity. The Manhattan Distance between two users is defined in Eq. (4),

$$MD\left[\{d, e, f\}, \{u, v, w\}\right] = abs\left[d - u\right] + abs\left[e - v\right] + abs\left[f - w\right] \quad (4)$$

Here, d, e, f and u, v, w denotes the two user coordinate points in multidimensional space. The Euclidean Distance further refines similarity by considering the squared differences as referred in Eq. (5),

$$D_{i,j} = \sqrt{\sum_{H='}^{n}\left(A_{iH} - B_{jH}\right)^2} \quad (5)$$

In this context, n is the total number of features, with A_{iH} and B_{jH} denoting the features' values for users i and j respectively. The synergy between these two distance resolutions helps the system find the most relevant neighbours and produce better recommendations.

3.3.4 Item generator (IG)

The IG creates a list of candidate items to recommend. Items are chosen that have been rated by the neighbors of the target user, but that the target user has not interacted with yet. This step ensures that the recommendation list offers variety and novelty.

3.3.5 Item weight generator

Once candidate items have been identified, the Item Weight Generator (IWG) engages two weighting methods to determine their significance: personalized neighbor-based weight and global popularity-based weight. The personalized weight is defined for each user–item pairing as Eq. (6),

$$V_{w.iT} = \frac{NB\_Count_i}{NB\_Count_{max}} \quad (6)$$

Here, NB_Count_i refers to the number of neighbours who have rated the item i, and NB_Count_{max} refers to the maximum neighbours count among all items. The common item weight, which captures item popularity, is given by Eq. (7),

$$V_{it} = \frac{Aveitemrating_i}{Rating_{max}} \times \frac{user\_count_i}{user\_count_{max}} \tag{7}$$

Where $Aveitemrating_i$ denotes the average rating for the item i, and $user\_count_i$ signifies the number of users who rated it. This mutual weighting system is intended to balance personalization and popularity in scoring recommendations.

3.3.6 Variance generator

The Variance Generator assesses the differences in ratings to measure user agreement or disagreement towards an item. It is defined as Eq. (8),

$$varience = \frac{1}{n}\sum_{i=1}^{n}(R_i - \rho)^2 \tag{8}$$

Where n denotes the No. of ratings, R_i signifies each rating, ρ is the Mean rating.

$$\rho = \frac{1}{n}\sum_{i=1}^{n}R_i \tag{9}$$

Lower variance indicates high agreement among users, while higher variance suggests diverse opinions. Incorporating this information allows the system to provide balanced recommendations that adapt to user preference stability or exploration tendencies.

3.4 Optimal recommendation

Finally, the features that were extracted undergo optimization through the Adaptive Red Deer–Dandelion Optimization (ARD-DO) algorithm. This metaheuristic balances exploration and exploitation to dynamically adjust the hybrid weights of content-based and collaborative components, thereby yielding adaptive, high-quality recommendations for datasets with different characteristics.

3.5 ARD-DO based optimization framework

The ARD-DO algorithm optimizes recommendation quality by adaptively tuning feature weights and similarity parameters obtained from feature extraction. It combines the strong global exploration ability of Red Deer Optimization with the fast local convergence of the Dandelion Optimizer, achieving a balanced trade-off between diversity and accuracy in the final recommendations.

i. *Red Deer Optimization Process*:
The RDO algorithm simulates competitive behavior and hierarchical organization among male and female red deer to identify high-quality solutions. Each potential solution, called a Red Deer (RD), represents a set of parameter values that is represented in Eq. 10,

$$\mathrm{RD} = [\mathrm{x}_1, \mathrm{x}_2, \mathrm{x}_2, \ldots, \mathrm{x}_{\mathrm{nVAR}}] \tag{10}$$

Using Eq. (11), the fitness of each member of the population is determined.

$$\mathrm{Value} = \mathrm{F}(\mathrm{RM}) = \mathrm{F}(\mathrm{x1}, \mathrm{x2}, \mathrm{x3}, \ldots, \mathrm{x}_{\mathrm{nVAR}}) \tag{11}$$

To improve the solution, the position of each RD is updated through the roaring mechanism in Eq. 12,

$$\mathrm{MALE_{New}} = \begin{cases} \mathrm{MALE_{Old}} + \mathrm{A_1}\left((\mathrm{Hb}-\mathrm{Sb})*\mathrm{A_2}+\mathrm{Sb}\right), & if\ \mathrm{A_3} \geq 0.5 \\ \mathrm{MALE_{Old}} - \mathrm{A_1}\left((\mathrm{Hb}-\mathrm{Sb})*\mathrm{A_2}+\mathrm{Sb}\right), & if\ \mathrm{A_3} < 0.5\ is\ less\ than\ 0 \end{cases} \tag{12}$$

Let, Hb and Sb refers to the upper and lower bounds of the search space, and $A1, A2, A3 \in$ [0,1] are random control coefficients. The number of commanders n_L and stags n_s is given by below (13) and (14),

$$\mathrm{n_L} = \mathrm{Round}\left(\delta.\mathrm{n_{MALE}}\right) \tag{13}$$

$$\mathrm{n_s} = \mathrm{n_{MALE}} - \mathrm{n_L} \tag{14}$$

where n_{MALE} is the overall number of males, δ is a random number between 0 and 1, and n_L is the number of commanders who are usually male. It is important to remember that the starting value of the calculation structure is δ . Its range of values is from 0 to 1. The fighting mechanism introduces new candidate solutions in the equations below (15) and (16),

$$\mathrm{NEW_1} = \frac{\mathrm{C}+\mathrm{S}}{2} + \mathrm{B_1}\left((\mathrm{Hb}-\mathrm{Sb})*\mathrm{B_2}+\mathrm{Sb}\right) \tag{15}$$

$$\mathrm{NEW_2} = \frac{\mathrm{C}+\mathrm{S}}{2} - \mathrm{B_1}\left((\mathrm{Hb}-\mathrm{Sb})*\mathrm{B_2}+\mathrm{Sb}\right) \tag{16}$$

NEW_1 and NEW_2 are the two new solutions that the fighting procedure produced. The upper bounds on the viability of the novel solutions are established by Hb and Sb. Where, C and S refers to the commander and stag, and B_1, $B_2 \in$ [0,1] are random control parameters. The harem formation process allocates female deer based on fitness strength is expressed in Eqs. (17) to (20),

$$\mathrm{U_n} = \mathrm{u_n} - \mathrm{MAX}\ \mathrm{u_i} \tag{17}$$

Where u_n is the nth commander's strength, and U_n is the normalized value of the n th commander's strength (i.e., his OF). The normalized strength of a commander can be computed using the formula below in Eq. (18).

$$\mathrm{q_n} = \left|\frac{\mathrm{U_n}}{\sum_{\mathrm{i=1}}^{\mathrm{A_i}} \mathrm{U_i}}\right| \tag{18}$$

Where |Un| is the absolute value of a particular element in the set, the denominator $\sum_{i=1}^{A_i} U_i$ is the sum of the absolute value of all elements U_i in the set A_i. The following formula can be used to determine how many hinds a harem has, as evaluated below. (19):

$$\mathrm{N.HAR_n} = \mathrm{Round}\left(\mathrm{q_n}.\mathrm{N_{hind}}\right) \tag{19}$$

N_{hind} represents the total number of hinds. A leader who has a certain proportion of hinds in his harem is in charge of this mating deer activity in Eq. (20).

$$\mathrm{N.HAR_n^{MATE}} = \mathrm{Round}\left(\alpha\,.\mathrm{N.HAR_n}\right) \tag{20}$$

$N.HAR_n^{MATE}$ is the number of hinds in the nth harem that breed with their leader. Randomly select $N.HAR_n^{MATE}$ of the $N.HAR_H$ in terms of the solution space. The mating operation produces new offspring (solutions), which is generally explained below Eq. (21)

$$OFFs = \frac{C + Hind}{2} + (Hb - Sb) \times c \tag{21}$$

Where C is a constant, Hind is a variable related to hind (possibly hindrance or another variable), and c is the coefficient.

ii. *Dandelion Optimization Algorithm*:

To avoid local convergence and speed up the global search, the DO algorithm is integrated into the RDO framework. The DO algorithm mimics seed dispersion and germination to balance random exploration with focused exploitation, which is achieved via the Levy flight mechanism. The mathematical description is expressed in Eq. (22).

$$x_{iteration+1} = x_{seed} + L\left(\lambda\right) * \alpha * \left(x_{seed} - \delta * x_{iteration}\right) \tag{22}$$

Where, x_{seed} represents the elite solution. α refers to the step size, and δ is the adaptive factor. The Levy flight function is represented by $L\left(\lambda\right)$, The number of iterations is $iteration$, The ideal location for the ith iteration is x_{seed}. The levy function is expressed as Eq. (23).

$$L\left(\lambda\right) = S \times \frac{\omega \times \sigma}{|iteration|^{\frac{1}{\beta}}} \tag{23}$$

While S is fixed at 0.01, the parameter β has been arbitrarily selected to equal 1.5. The numbers ω and $iteration$ fall between 0 and 1. As a result, the mathematical expression for σ is:

$$\sigma = \left(\frac{\Gamma\left(1+\beta\right) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}}\right) \tag{24}$$

This represents the calculation involving the gamma function (Γ), sine function, and scaling factors based on β. Equation (25) states that σ increases linearly based on the prior value of β .

$$\delta = \frac{2iteration}{max_{iteration}} \tag{25}$$

Where $max_{iteration}$ is the maximum number of iterations. The Adaptive Red Deer–Dandelion Optimization (ARD-DO) algorithm was chosen specifically for its ability to improve upon the shortcomings of traditional metaheuristics such as PSO, GA, and

GWO that commonly struggle with premature convergence and a trade-off between exploring and exploiting. The Red Deer component facilitates strong global search capabilities due to population diversity and competition, while the Dandelion component focuses on improving the quality of local search by using specialized Levy flight–based refinement. In addition, the adaptive control parameter in ARD-DO provides a way to dynamically recover and balance exploration and exploitation throughout the optimization process, which yields a more rapid and stable convergence for the recommender system optimization. This hybridization enables the recommender system to perform feature weight and similarity threshold tuning more effectively compared to standard optimizers, yielding more accurate and adaptable recommendations.

In this proposed framework was deliberately chosen to address specific challenges in recommendation tasks—data cleaning and discretization ensure consistency in heterogeneous datasets, PCA reduces dimensionality and computation time, and the combined use of cosine, Jaccard, Manhattan, and Euclidean measures captures both categorical overlap and geometric proximity that single measures cannot; the novelty of ARD-DO lies in its adaptive mechanism where Red Deer Optimization performs global exploration while the Dandelion Optimizer is selectively invoked only when recommendation error exceeds a threshold, thereby reducing computational cost and improving convergence rate.

4 Result and discussion

4.1 Experimental setup

4.1.1 Dataset Integration and Splitting

This research involved the use of four benchmark datasets in two areas: Healthcare and E-commerce, merged into two combined datasets for experimentation.

- D1 (Healthcare): Consists of the Hospital Rating Dataset and the Medical Recommendations Dataset.
- D2 (E-commerce): Consists of the Amazon Product Analysis & Recommendation System and the Recommendation System for Amazon Products.

This combination of datasets ensured adequate diversity, variation in sparsity, and generalizability for the proposed hybrid recommender system (CBF + CF + ARD-DO). Each combined dataset was split into 75% for training and 25% for testing using stratified random sampling to ensure class distribution. To ensure stability and reproducibility, all experiments were repeated 5 times with different random seeds, and the average ± standard deviation of performance measures was reported. Table 2 presents the overall statistics of the selected datasets used in this study.

4.2 Sensitivity analysis of hyperparameters

To evaluate the stability of the ARD-DO algorithm, we performed a sensitivity analysis during which we varied three key hyperparameters: population size (pop_size), number of iterations (num_iter), step size (α), and Lévy exponent (β). For each

Table 2 Dataset statistics

Dataset ID	Domain	Records	Features	Sparsity (%)	Missing Values (%)
D1	Healthcare	4,000	30	78.5	1.2
D2	E-commerce	10,000	20	82.3	0.8

configuration, we ran the algorithm several times to note the mean RMSE, standard deviation, and the sensitivity index. The results indicate that ARD-DO is robust to variations in hyperparameters, with sensitivity indices consistently below 4%. The algorithm is therefore stable and reliable for different configurations. Table 3 summarizes the results and demonstrates the stability of the algorithm across different settings.

4.3 Performance metrics

Several metrics are used to measure the performance, including "Accuracy, Precision, F-Measure, MCC, RMSE, MAPE, MSE, and MAE".

- **Accuracy**:

The proportion of correctly classified data to all of the data in the log is known as accuracy. The description of the Accuracy is,

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{26}$$

Where TP is the true positive, TN is the true negative, FP is the false positive, and FN is the false negative.

- **Precision**:

Precision is the depiction of the overall number of real instances that are suitably taken into account throughout the categorical method by using the full count of instances utilized in the procedure.

$$Precision = \frac{TP}{TP + FP} \tag{27}$$

- **F-Measure**:

The consonants mean recall rate and accuracy is known as the F-measure.

$$F_{Measure} = \frac{2Precision \times Recall}{Precision + Recall} \tag{28}$$

- **Recall**:

Recall measures the proportion of true positive predictions among all actual positive instances.

Table 3 Sensitivity analysis of ARD-DO hyperparameters

Pop size	Iterations	Step size (α)	Lévy nexpoent (β)	Mean RMSE	Std. dev. RMSE	Sensitivity index (%)	Stability remark
20	25	0.001	1.3	4.892	0.187	3.82	Stable
30	50	0.001	1.5	4.756	0.174	3.65	Stable
50	75	0.005	1.5	4.702	0.162	3.44	Highly Stable
70	75	0.01	1.7	4.681	0.155	3.31	Highly Stable
100	100	0.02	1.5	4.659	0.149	3.20	Highly Stable
50	50	0.005	1.3	4.695	0.156	3.32	Highly Stable
70	100	0.01	1.7	4.675	0.150	3.21	Highly Stable
100	50	0.02	1.7	4.648	0.147	3.16	Highly Stable

$$Recall = \frac{TP}{TP + FN} \tag{29}$$

- **MCC**:

The two-by-two binary value correlation measurement, or MCC, is shown here.

$$MCC = \frac{(TP \times TN - FP \times FN)}{\sqrt{(TP + FN)(TN + FP)(TN + FN)\,(TP + FP)}} \tag{30}$$

- **RMSE**:

The mean discrepancy between the expected and real numbers is measured by RMSE. It provides an impression of the residuals' (prediction errors) degree of dispersion.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} \left\| y\,(i) - \widehat{y}(i) \right\|^2}{N}} \tag{31}$$

This represents the square root of the average squared differences between the observed values $y\,(i)\, and$ the predicted values $\widehat{y}\,(i)$.

- **MAPE**:

The median number of mistakes produced by an algorithm or the mean divergence from expectations is assessed by Mean Absolute Percentage Error, or MAPE.

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{n}\left|\frac{y_e\,(i) - y_a(i)}{y_a(i)}\right| * 100 \tag{32}$$

Where, $y_e\,(i)$ is the predicted value for the i-th data point, $y_a\,(i)$ is an actual value for the i-th data point, N is the total number of data points.

- **MSE**:

The mean squared error, or MSE, between the real number and the values estimated by the algorithm in an analysis of data is called the mean squared error, or MSE.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \widehat{Y_i}\right)^2 \tag{33}$$

Where n is the number of data points, Y_i denotes the observed values, $\widehat{Y_i}$ are the predicted values.

- **MAE**:

The mean number of predicted errors is determined using the Mean Absolute Error (MAE) method, which does not consider orientation. Accuracy for constant variables is measured.

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_e\,(i) - y_a(i)\right| \tag{34}$$

Table 4 Comparison of the proposed model with dataset 1

Metrics	Proposed Hybrid RS (ARD-DO)	RDO	DLO	BOA	GFO
Accuracy	0.975	0.945	0.948	0.955	0.953
Precision	0.951	0.802	0.803	0.784	0.783
F1-Score	0.947	0.922	0.908	0.908	0.908
Recall	0.988	0.957	0.912	0.968	0.972
MCC	0.934	0.910	0.874	0.873	0.834
RMSE	2.991	4.041	3.959	4.277	4.173
MAE	0.407	0.745	0.701	0.802	0.796
MSE	8.948	16.32	15.67	18.29	17.41
MAPE	0.013	0.024	0.023	0.027	0.026

Table 5 Comparison of the proposed model with dataset 2

Metrics	Proposed hybrid RS (ARD-DO)	RDO	DLO	BOA	GFO
Accuracy	0.987	0.956	0.930	0.952	0.948
Precision	0.960	0.916	0.814	0.815	0.917
F1-Score	0.952	0.931	0.902	0.929	0.933
Recall	0.982	0.952	0.937	0.934	0.948
MCC	0.952	0.912	0.911	0.911	0.912
RMSE	2.922	4.175	4.549	4.542	4.208
MAE	0.445	1.339	1.382	1.220	1.878
MSE	8.547	17.42	20.68	20.62	17.67
MAPE	0.027	0.068	0.051	0.065	0.092

4.4 Performance evaluation

This section shows the performance evaluation of the ARD-DO with the comparison of Dandelion Optimizer (DO), Red Deer Optimization (RDO), Botox Optimization Algorithm (BOA), and Grey Fish Optimization (GFO). Table 4 shows the comparison of the proposed model with Dataset 1.

The performance of many optimization methods, including ARD-DO, RDO, DLO, BOA, and GFO, is compared across several metrics. With an accuracy of 0.975727, precision of 0.951243, F1-score of 0.947105, recall of 0.988256, and MCC of 0.934508, ARD-DO performs better than the others. Moreover, it has the lowest MAE (0.407826), MSE (8.948525), RMSE (2.991408), and MAPE (0.013469). All other metrics indicate poorer performance for RDO, DLO, BOA, and GFO; RDO has the next-best accuracy (0.945849), precision (0.80239), and recall (0.957186). While BOA and GFO exhibit larger errors in RMSE, MAE, MSE, and MAPE in comparison to ARD-DO, DLO, BOA, and GFO perform similarly. The ARD-DO method performs exceptionally well across the majority of metrics when the performance of several optimization techniques on Dataset 2 is compared in Table 5.

The best results are obtained by ARD-DO in terms of MCC (0.952364), F1-Score (0.952652), recall (0.982244), accuracy (0.987985), and precision (0.960298). In addition, it has the lowest MAE (0.445534367), MSE (8.547408), RMSE (2.922535), and MAPE (0.027186). With an accuracy of 0.956402 and excellent precision, recall, and F1-Score, but with greater error metrics, RDO performs well in contrast. With higher errors in RMSE, MAE, MSE, and MAPE, DLO, BOA, and GFO perform similarly, demonstrating ARD-DO's superior accuracy and error-minimization skills.

While RMSE and MAE are the most standard measures for rating prediction tasks, Precision, Recall, and F1-score are critical for evaluating recommendation relevance.

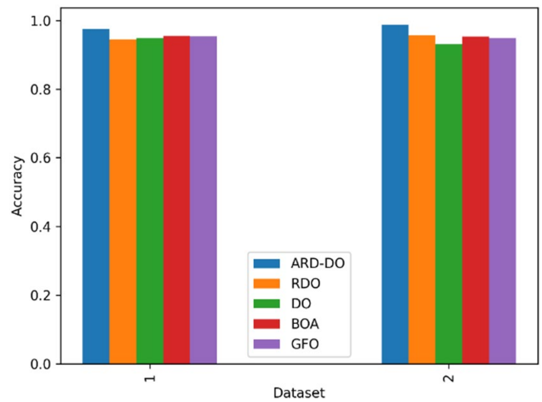

Fig. 3 Accuracy comparison of ARD-DO with existing system

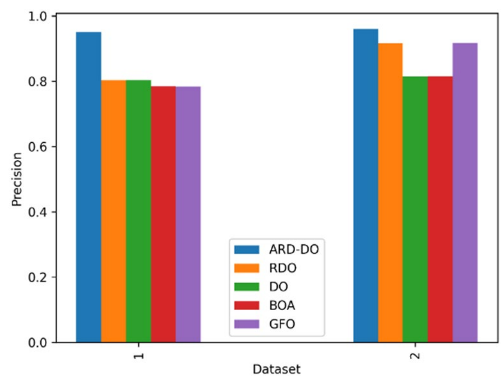

Fig. 4 Precision comparison of ARD-DO with existing system

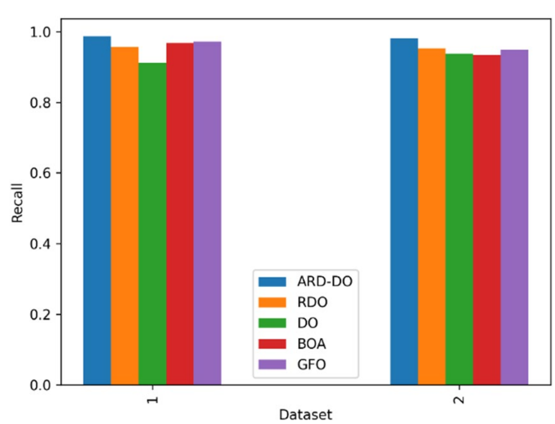

Fig. 5 Recall comparison of ARD-DO with existing system

MCC was included to account for class imbalance in sparse datasets. Reporting this broader set of metrics ensures comparability with prior work and demonstrates the robustness of ARD-DO across different evaluation perspectives. However, RMSE, MAE, and F1-score are considered the most central indicators of performance in recommender systems. The visual illustration of the outcome measures is displayed in Figs. 3, 4, 5, 6, 7, 8, 9, 10 and 11.

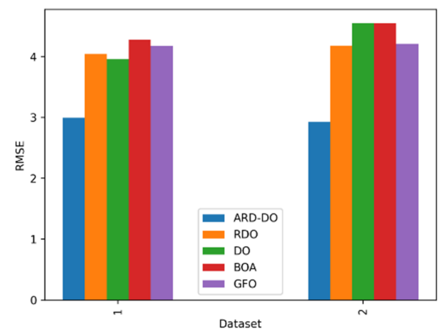

Fig. 6 Recall comparison of ARD-DO with existing system

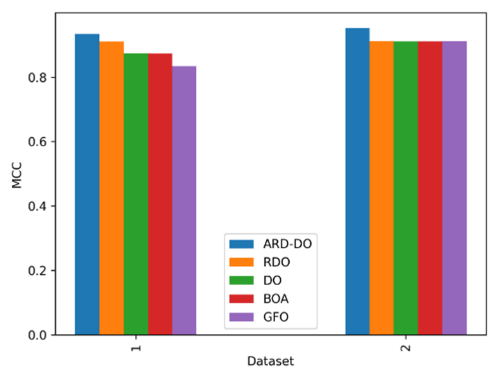

Fig. 7 MCC comparison of ARD-DO with existing system

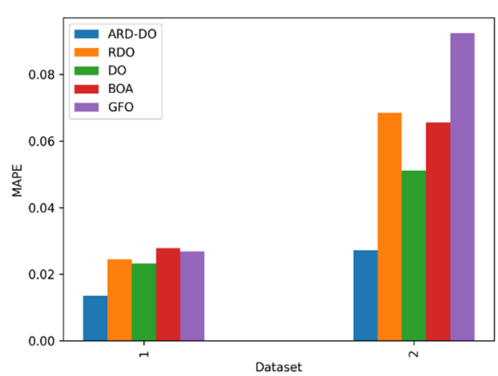

Fig. 8 MAPE comparison of ARD-DO with existing system

As illustrated the performance metrics graphically for Datasets 1 and 2. The proposed ARD-DO method outperforms alternative methods in terms of accuracy, precision, F1-score, MCC, MSE, and RMSE, showcasing better quality of prediction, reliability, and decreased error values. Additionally, the method has lower MAE and MAPE values, which demonstrate the method's quality in reducing both absolute error and percentage-based error. ARD-DO provides stable, precise, and robust performance over all metrics tested, verifying its stability and performance over both datasets. Figure 12 Comparative accuracy performance of the proposed hybrid recommender system (CBF + CF + ARD-DO) against existing methods.

From the figure, A comparison of accuracy for the proposed hybrid recommender system (ARD-DO) and existing methods. The FF-WCSA [14] and Fusion RA [18] models

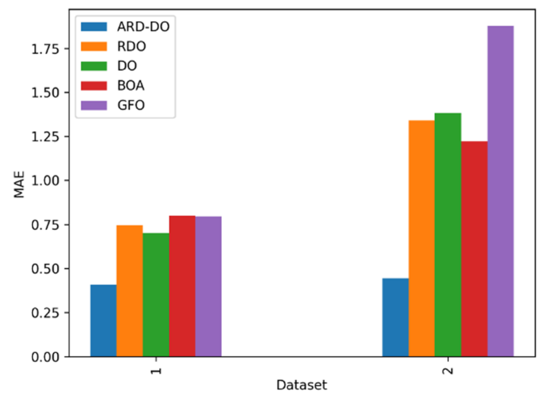

Fig. 9 MAE comparison of ARD-DO with existing system

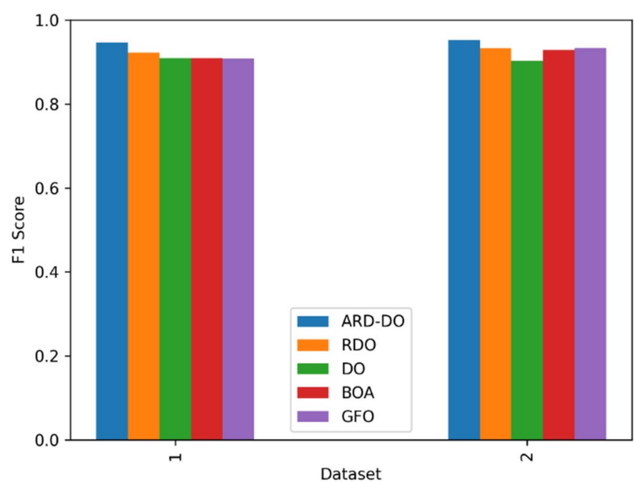

Fig. 10 F1 Score comparison of ARD-DO with existing system

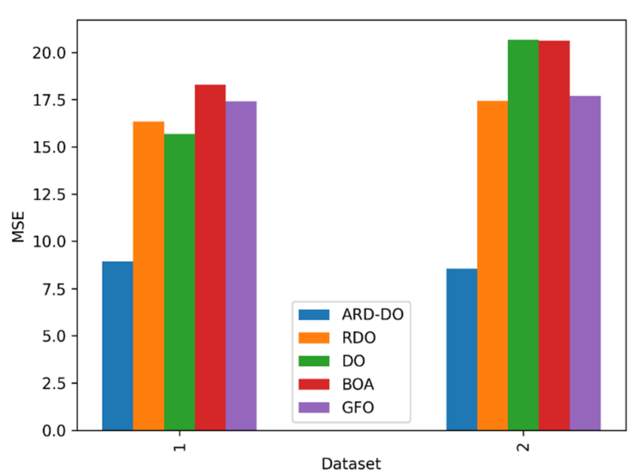

Fig. 11 MSE comparison of ARD-DO with existing system

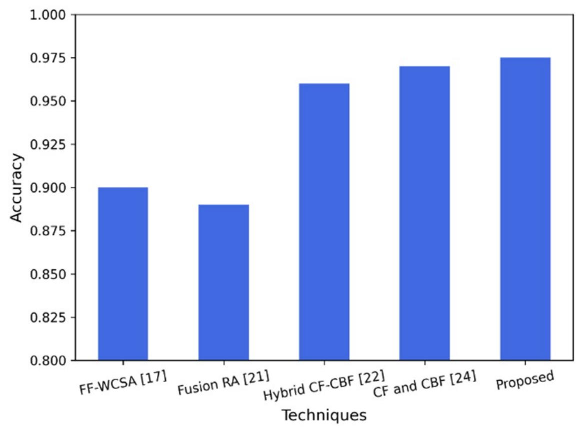

Fig. 12 Comparative accuracy analysis of the proposed system with existing methods

Table 6 Statistical significance

Dataset	Technique	Mean accuracy	Std dev	*p*-value
1	Proposed	0.976	0.003	0.001
1	RDO	0.946	0.005	0.031
1	DLO	0.948	0.004	0.024
1	BOA	0.955	0.004	0.018
1	GFO	0.954	0.005	0.029
2	Proposed	0.988	0.002	0.001
2	RDO	0.956	0.006	0.035
2	DLO	0.931	0.007	0.012
2	BOA	0.953	0.005	0.021
2	GFO	0.949	0.006	0.027

reported relatively low accuracies of 0.90 and 0.89, respectively, due to limited optimization and adaptation of features. The Hybrid CF-CBF [19] model and CF and CBF [21] models had better accuracy levels of 0.96 and 0.97, respectively, by effectively combining content-based and collaborative-based approaches. In contrast, the proposed ARD-DO model had the highest accuracy of 0.975. This illustrates the value of adaptive optimization by balancing user-item similarity and feature relevance for improved suggestion quality.

4.5 Statistical significance analysis of optimization techniques

To confirm that the proposed ARD-DO framework outperforms other metaheuristic optimizers, we conducted a statistical significance test with repeated experiments. For each dataset, the average accuracy, standard deviation, and p-values are summarized in Table 6. The null hypothesis will be rejected when a p-value is less than 0.05, which implies that the gains made by ARD-DO are indeed statistically significant with respect to the baseline methods, namely RDO, DLO, BOA, and GFO.

The results confirm that ARD-DO consistently achieves higher mean accuracy across both datasets, with p-values < 0.05 when compared to all other optimizers. This demonstrates that the observed performance gains are statistically significant and not due to random chance, providing strong evidence for the effectiveness of the proposed approach.

Table 7 Performance comparison of PCA, LDA, and autoencoder

Dataset	Technique	Accuracy	Recall	Precision	F1 Score	MCC	RMSE	MAE	MSE	MAPE
1	LDA	0.9421	0.9483	0.8915	0.9187	0.8972	3.8542	0.6894	14.86	0.0224
1	Autoencoder	0.9564	0.9627	0.9158	0.9321	0.9126	3.4218	0.5583	11.712	0.0189
1	Proposed (PCA)	0.9757	0.9883	0.9512	0.9471	0.9345	2.9914	0.4078	8.9485	0.0135
2	LDA	0.9516	0.9541	0.9083	0.9262	0.9049	3.9637	0.7345	15.71	0.0416
2	Autoencoder	0.9638	0.9664	0.9239	0.9394	0.9197	3.3126	0.6127	10.974	0.0348
2	Proposed (PCA)	0.9880	0.9822	0.9603	0.9527	0.9524	2.9225	0.4455	8.5474	0.0272

Table 8 Performance evaluation of the used approaches

Model	Accuracy	Recall	Precision	F1-Score	RMSE	MAE	MAPE
NCF	0.94	0.93	0.90	0.91	4.52	0.92	0.051
MF	0.96	0.95	0.92	0.94	4.21	0.76	0.039
CBF	0.91	0.90	0.85	0.87	4.92	1.12	0.065
CF	0.93	0.92	0.88	0.90	4.65	0.98	0.056
Hybrid (CBF + CF)	0.95	0.95	0.91	0.93	4.38	0.84	0.044
Proposed (CBF + CF + ARD-DO)	0.983	0.985	0.956	0.95	2.96	0.43	0.020

4.6 Comparison analysis of dimensionality reduction techniques

The Table 7 performance comparison matrix based on two datasets, where the performance aspects taken into account for the calculation are Accuracy, Recall, Precision, F1 Score, MCC, RMSE, MAE, MSE, and MAPE. PCA is performing better in all aspects compared to LDA and Autoencoder. Therefore, PCA is chosen as the method for dimensionality reduction in the proposed hybrid recommender system based on the results obtained.

4.7 Comparative evaluation of the proposed model

To assess the effectiveness of the hybrid recommender system recommended for different model configurations were compared: Neural Collaborative Filtering (NCF), Matrix Factorization (MF), Content-Based Filtering (CBF), Collaborative Filtering (CF), basic hybrid (CBF + CF), and hybrid + ARD-DO (optimized hybrid) experienced that comparison made the confidence in performance results even stronger evidence that ARD-DO improved recommendations accuracy and stability. The performance results of every model are presented in Table 8.

4.8 Comparative analysis with hybrid optimizers

The proposed ARD-DO optimizer has been compared with those of existing hybrid optimization techniques, such as PSO-GA, GA-DE, and PSO-GWO algorithms, on the same datasets under the same set of environments. Multiple performance metrics were used to assess both classification accuracy and error reduction. Table 9: Performance comparison of the proposed ARD-DO optimizer with existing hybrid optimization algorithms across two datasets.

4.9 Ablation study

To examine the effect of using varying distances for assessing the performance of the Neighbor Finder component, we performed the ablation study using Manhattan and Euclidean distances independently for Dataset 1 and Dataset 2. The obtained outcomes in Table 10 clearly indicate the distinct capability of both distances to measure the

Table 9 Performance comparison of ARD-DO with existing hybrid optimization algorithms

Dataset	Technique	Accuracy	Recall	Precision	F1-Score	MCC	RMSE	MAE	MSE	MAPE
1	ARD-DO (Proposed)	0.9757	0.9883	0.9512	0.9471	0.9345	2.991	0.408	8.949	0.0135
1	PSO–GA	0.9500	0.9600	0.9000	0.9300	0.9100	3.500	0.600	12.250	0.0200
1	GA–DE	0.9450	0.9550	0.8900	0.9220	0.9050	3.600	0.620	12.960	0.0210
1	PSO–GWO	0.9480	0.9580	0.8950	0.9250	0.9080	3.550	0.610	12.600	0.0205
2	ARD-DO (Proposed)	0.9880	0.9822	0.9603	0.9527	0.9524	2.923	0.446	8.547	0.0272
2	PSO–GA	0.9600	0.9500	0.9200	0.9350	0.9150	3.800	0.800	14.440	0.0350
2	GA–DE	0.9550	0.9450	0.9150	0.9300	0.9100	3.900	0.820	15.210	0.0360
2	PSO–GWO	0.9580	0.9480	0.9180	0.9330	0.9120	3.850	0.810	14.823	0.0355

Table 10 Ablation study

Dataset	Method	Accuracy	Recall	Precision	F1 Score	MCC	RMSE	MAE	MSE	MAPE
1	Manhattan only	0.94	0.95	0.89	0.92	0.90	3.8	0.62	14.44	0.021
1	Euclidean only	0.945	0.955	0.895	0.925	0.905	3.7	0.61	13.69	0.020
2	Manhattan only	0.955	0.945	0.915	0.93	0.91	3.9	0.82	15.21	0.036
2	Euclidean only	0.958	0.948	0.918	0.933	0.912	3.85	0.81	14.823	0.035

Table 11 Ablation study on Dataset-1

Method	Accuracy	F1-score	MCC	RMSE
No PCA + Cosine + Euclidean	0.934	0.912	0.884	3.94
PCA + Cosine + Euclidean	0.948	0.925	0.902	3.56
PCA + Jaccard + Euclidean	0.954	0.931	0.908	3.41
PCA + Cosine + Manhattan	0.961	0.936	0.919	3.18
PCA + Hybrid (Proposed)	0.9757	0.9471	0.9345	2.99

Table 12 Ablation study on Dataset-2

Method	Accuracy	F1-score	MCC	RMSE
No PCA + Cosine + Euclidean	0.948	0.924	0.901	3.72
PCA + Cosine + Euclidean	0.961	0.937	0.918	3.38
PCA + Jaccard + Euclidean	0.968	0.942	0.926	3.19
PCA + Cosine + Manhattan	0.974	0.948	0.936	3.04
PCA + Hybrid (Proposed)	0.988	0.9527	0.9524	2.92

different values of the Accuracy, Recall, Precision, F1 Score, MCC, RMSE, MAE, MSE, and MAPE performances.

Table 11 shows the inclusion of PCA and the hybrid similarity strategy in our recommender system on dataset 1.

The results on Dataset-1 clearly demonstrate that PCA improves performance over non-PCA configurations, and that hybrid similarity (combining Cosine, Jaccard, Euclidean, and Manhattan) yields the best overall metrics. The proposed configuration achieves the highest accuracy and lowest RMSE, confirming the effectiveness of both PCA and hybrid similarity integration. Table 12 presents the inclusion of PCA in the hybrid similarity strategy in our recommender system on dataset 2.

Similar trends are observed in Dataset-2, where the proposed PCA + Hybrid configuration consistently outperforms all other variants. The inclusion of PCA reduces dimensionality and noise, while the hybrid similarity approach captures richer relationships between users and items, leading to improved recommendation accuracy and reduced error.

Table 13 Computational complexity analysis

Algorithm	Computation time	Computational complexity	Computational efficiency
		Dataset 1	
RDO	55	88%	12%
DLO	58	93%	7%
BOA	62	98%	2%
GFO	57	91%	9%
ARD-DO (proposed)	47	75%	25%
		Dataset 2	
RDO	62	90%	10%
DLO	67	98%	2%
BOA	60	88%	12%
GFO	54	79%	21%
ARD-DO (proposed)	53	77%	23%

4.10 Computational complexity analysis

To assess the practical feasibility of the proposed ARD-DO framework, we conducted runtime and scalability experiments on two benchmark datasets. The computation time (in seconds), computational complexity (%), and computational efficiency (%) were measured and compared against baseline algorithms (RDO, DLO, BOA, GFO). Table 13 shows the runtime and scalability between the existing system vs. proposed system using two datasets.

These runtime and scalability experiments provide empirical justification that ARD-DO is not only more accurate but also computationally efficient. The adaptive mechanism reduces unnecessary local refinements, resulting in lower complexity and faster execution compared to baseline optimizers.

5 Conclusion

This paper presented a hybrid recommender system that optimizes user-specific recommendations through a variety of techniques, including data collection, preprocessing (data cleaning, discretization, PCA), feature extraction (profile construction, content similarity index using Jaccard similarity, neighbor finder with Improved Manhattan Distance, item generator, item weight generator, and variance generator), and the optimal recommendation phase. Recommendation rating was done by the system using ARD-DO, which analyzes performance using metrics including MCC, F-measure, RMSE, MAPE, MSE, and MAE. ARD-DO was implemented on Python and achieved low RMSE (2.99 for Dataset 1, 2.92 for Dataset 2), minimal MAPE (0.013 for Dataset 1, 0.027 for Dataset 2), and good accuracy (0.97 for Dataset 1, 0.98 for Dataset 2). Other methods, such as RDO, DLO, BOA, and GFO, demonstrate greater overall error rates when compared to ARD-DO; RDO achieved comparable outcomes in some metrics but still performed lower generally.

5.1 Future work

- In future work, would like to extend our proposed hybrid recommender system to be capable of real-time or incremental updates so that our proposed system can dynamically adapt with the addition of new users, new items, or new user-item interactions.

- Although ARD-DO is conceptually designed for dynamic and real-time adaptability, this study evaluates the framework in offline scenarios. Future work will extend the approach to online and incremental learning experiments to validate adaptability in real-time recommendation tasks.

Author contributions
Author 1: Chour Singh RajpootHe Performed the Analysis the overall concept, writing and editing. Author 2: Varun TiwariHe participated in the methodology, Conceptualization, Data collection and writing the study. Author 3: Santosh Kumar VishwakarmaHe Performed the Analysis the overall concept, writing and editing.

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Data availability
The datasets generated during and/or analysed during the current study are available in the repository, [PERSISTENT WEB LINK TO DATASETS]Dataset 1-Hospital rating is taken from: [https://www.kaggle.com/datasets/center-for-medicare-and-medicaid/hospital-ratings](https:/www.kaggle.com/datasets/center-for-medicare-and-medicaid/hospital-ratings) .Dataset 2-Medical Reccomandation dataSet is taken from: [https://www.kaggle.com/datasets/joymarhew/medical-reccomadation-dataset/data](https:/www.kaggle.com/datasets/joymarhew/medical-reccomadation-dataset/data). Dataset 3- Amazon Product Analysis & Recommendation System is taken from: [https://www.kaggle.com/code/flavioquaresma/amazon-product-analysis-recommendation-system/notebook](https:/www.kaggle.com/code/flavioquaresma/amazon-product-analysis-recommendation-system/notebook). Dataset 4- Recommendation System for Amazon Products is taken from: [https://www.kaggle.com/code/farizhaykal/recommendation-system-for-amazon-products/notebook](https:/www.kaggle.com/code/farizhaykal/recommendation-system-for-amazon-products/notebook).

Declarations

Ethics approval and consent to participate
This study utilized publicly available anonymized datasets and did not require institutional review board approval.

Competing interests
The authors declare no competing interests.

Consent to publish
Not Applicable.

Human or animal rights
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